

Date

Date

Descriptive Statistics

Measure of Central Tendency:

- ⇒ We uses:
- a) Mean
 - b) Median
 - c) Mode

⇒ Mean:

Popⁿ mean (μ)

Sample Mean ($\bar{x}$)

$$\mu = \frac{\sum_{i=1}^N x_i}{N}$$

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

where, x is a random variable

⇒ charo. of Mean:

- ① affected by extreme outliers
- ② used for interval & ratio data

⇒ Median: middle value in a data set (arranging/data in order)

→ not get much impacted by outliers

⇒ char. of Median:

① not much impacted by outliers

- ② used for ordinal interval & ratio data

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⇒ Mode: value that appears most frequently in a dataset.

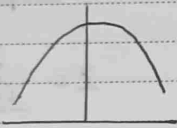
⇒ char. of Mode:

① Not affected by outliers

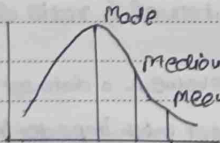
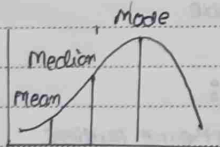
② used for Nominal, ordinal, interval & ratio data.

⇒ Important: (when to use what?)

① Mean: when data is symmetrically distributed without outliers.



② Median: when data is skewed or contain outliers.



for most common choice

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③ Mode: used for categorical data

to identify the most common category.

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⇒ Real world appⁿ:

① Feature engineering

Measure of Dispersion:

⇒ ways: ① Range ② Variance ③ Standard deviation

④ IQR (Interquartile Range)

⇒ Range: Max. value - Min. value

⇒ char. of Range:

① simple to calc.

② sensitive to outliers

⇒ Variance: It measures the avg. sq. deviation of each value from the mean. It provides a sense of how much the value in a dataset vary.

Popⁿ variance (σ^2)

Sample variance (s^2)

$$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$$

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$$

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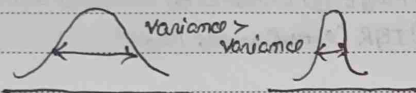
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Popⁿ size

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⇒ Char. of Variance:

- (a) Precise measure of variability
- (b) units are squared
- (c) sensitive to outliers than the Range
- (d) Variance dete. Spread of data



⇒ Std. deviation: sq. root of variance

$$s = \sqrt{\frac{\sum_{i=1}^N (x_i - \bar{x})^2}{N}}$$

⇒ Char. of Std. deviation:

- (a) sensitive to outliers
- (b) clear measure of spread in same unit

Why sample variance is divided by $n-1$:

⇒ To don't underestimate the true popⁿ variance. This correction is actually called as Bessel's correction.

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Random Variables:

⇒ It's a fun where values are derived from different processes or experiments.

e.g.

Tossing a coin → Head (0) or Tail (1)

$$X = \begin{cases} 0 \\ 1 \end{cases}$$

⇒ Types of random var:

(a) discrete

e.g., rolling a die,
tossing a coin

(b) continuous

e.g., inches of rain

Percentile and Quantiles:

⇒ Percentage:

{1, 2, 3, 4, 5, 6}

∴ No. of odd nos: 3

$$\therefore \% \text{ of " " } = \frac{3}{6} \times 100 = 50\%$$

⇒ Percentile: Value below which a certain % of observations lie.
e.g., {2, 2, 3, 4, 5, 5, 6, 7, 8, 8, 9, 9, 10}

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∴ Percentile of value 9 = $\frac{\text{no. of value below 9}}{n} \times 100$

$$= \frac{11}{14} \times 100 = 78.57 \text{ percent of value 9}$$

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Also, 25 percentile from distⁿ:

$$\text{value} = \frac{\text{Percentile} \times (n+1)}{100} = \frac{25 \times 15}{100} = 3.75$$

$$\therefore \frac{3+4}{2} = 3.5$$

$$\therefore 3.75 \approx 3.5$$

⇒ Quartile:

25 percentile = 1st Quartile

50 " = 2nd "

75 " = 4th "

5 No. Summary:

- ⇒ ① Minimum
- ② 1st Quartile
- ③ Median
- ④ 3rd Quartile
- ⑤ Maximum

$$\Rightarrow \text{Lower fence} = Q_1 - 1.5(IQR)$$

$$\text{Upper fence} = Q_3 + 1.5(IQR) \rightarrow \text{above it is outlier}$$

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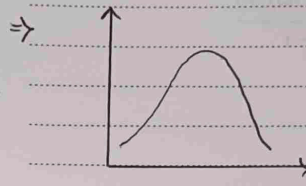
$$\Rightarrow IQR = Q_3 - Q_1$$

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Histogram and Skewness:

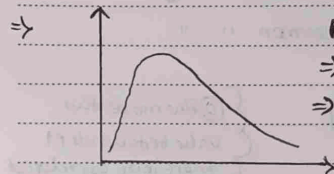


Normal / Gaussian distⁿ

⇒ symmetrical distⁿ

⇒ unskewness

⇒ Mean, median & mode exact at the center.

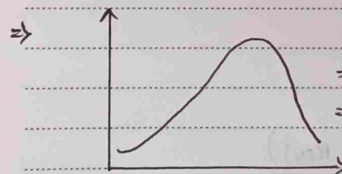
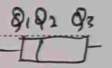


Right skewed distⁿ

⇒ +ve skewed

⇒ log normal distⁿ

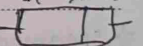
⇒ Mean > Median > mode



Left skewed distⁿ

⇒ -ve skewed

⇒ Mode > Median > mean



used to identify the relationship b/w two variables

Covariance and Correlation:

⇒ Covariance: It is a measure of how much two random

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variables change together.

⇒ +ve covariance: X↑ Y↑

-ve " : X↓ Y↑

→ disadv: ⑥ It doesn't have a specific limit value

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⇒ Covariance:

$$\text{Cov}(x, y) = \sum_{i=1}^n \frac{(x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

Also, $\text{Cov}(x, x) = \text{Var}(x)$

⇒ Correlation:

→ Pearson Corr Coeff

→ Spearman " "

⇒ Pearson Corr Coeff: $[-1, 1]$

⑥ The more the value be towards +1 more it be correlated

$$\rho_{x,y} = \frac{\text{Cov}(x, y)}{\sigma_x \cdot \sigma_y}$$

⇒ Spearman Rank Corr:

$$\gamma_s = \frac{\text{Cov}(R(x), R(y))}{\sigma(R(x)) \cdot \sigma(R(y))}$$



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